

Hardware Implementation of Arithmetic Units – Addition and Subtraction of Signed Numbers

1. Introduction

An **Arithmetic Unit (AU)** is a major component of the **Arithmetic Logic Unit (ALU)** in a computer system. It performs arithmetic operations such as **addition, subtraction, multiplication, and division**.

In digital computers, numbers are represented in **binary form**. Arithmetic operations on signed numbers require special representation methods such as **sign-magnitude, 1's complement, and 2's complement**.

Among these, **2's complement representation** is widely used because it simplifies the hardware design for addition and subtraction.

2. Representation of Signed Numbers

A **signed number** consists of two parts:

Bit	Meaning
MSB (Most Significant Bit)	Sign bit
Remaining bits	Magnitude

Sign Bit	Meaning
0	Positive
1	Negative

Example (8-bit representation)

+7 → 00000111

-7 → 11111001 (2's complement form)

3. Hardware Implementation of Arithmetic Unit

The hardware for arithmetic operations mainly consists of:

1. **Binary Adder**
 2. **Complement Circuit**
 3. **Multiplexers**
 4. **Registers**
 5. **Control Logic**
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4. Addition of Signed Numbers

Addition of signed numbers depends on the **signs of the operands**.

Case 1: Same Sign Addition

If both numbers have the **same sign**, their magnitudes are added and the result keeps the same sign.

Example:

+7 + +5

00000111
+00000101

00001100 = +12

Example:

-7 + -5

11111001
+11111011

11110100 = -12

Overflow Condition

Overflow occurs when:

- Two **positive numbers produce a negative result**, or
- Two **negative numbers produce a positive result**

Example:

0111 (+7)
+0100 (+4)

1011 (overflow)

Overflow is detected using **carry into MSB and carry out of MSB**.

5. Subtraction of Signed Numbers

In digital systems, subtraction is performed using **2's complement addition**.

Rule

$$A - B = A + (2's \text{ complement of } B)$$

Steps:

1. Find **2's complement** of the subtrahend.
 2. Add it to the minuend.
 3. Ignore final carry if present.
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Example 1

Calculate:

$$9 - 5$$

Binary:

9 = 00001001

5 = 00000101

2's complement of 5:

00000101

11111010 (1's complement)

+00000001

11111011

Add:

00001001
+11111011

00000100

Result = 4

Example 2 (Negative Result)

5 – 9

Binary:

5 = 00000101

9 = 00001001

2's complement of 9:

00001001

11110110

+1

11110111

Add:

00000101

+11110111

11111100

Result = -4

6. Hardware Implementation of Adder-Subtractor

A **single hardware circuit** can perform both addition and subtraction.

Components used:

- **n-bit Binary Adder**
- **XOR gates**
- **Control signal (M)**

Operation

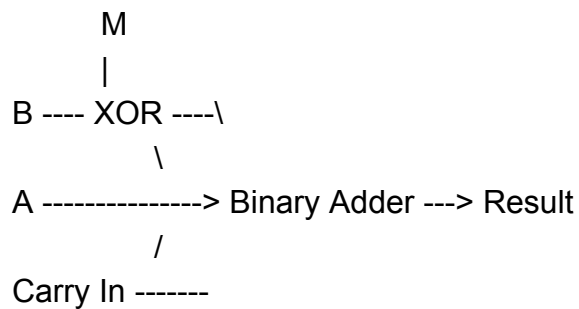
Control Signal (M)	Operation
0	Addition
1	Subtraction

Working

- When **M = 0**
 - B passes unchanged
 - Operation: **A + B**
- When **M = 1**
 - B is complemented using XOR gates
 - Carry input = 1
 - Operation becomes **A + 2's complement of B**

Thus subtraction is performed.

Hardware Logic



If $M = 0 \rightarrow$ Add

If $M = 1 \rightarrow$ Subtract

7. Advantages of Using 2's Complement

1. Same hardware used for **addition and subtraction**

2. Simplifies arithmetic circuit design
 3. Only **one zero representation**
 4. Efficient overflow detection
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8. Applications

Arithmetic units are used in:

- **Microprocessors**
- **Digital Signal Processors**
- **Embedded systems**
- **Scientific computing systems**